

The Convergence Atlas — Field Guide

Not just a beautiful diagram. Every point, colour, thread and number is data. This guide shows you how to read it, and how to make it reveal a hidden mechanistic link between two rare diseases.

01 The anatomy

Four elements, one idea: diseases on the outside, the mechanisms they share on the inside, and threads connecting them.



The whole Atlas at a glance.

The ring — every rare disease, grouped into 21 colour-coded clinical families (RASopathy, Lysosomal, Neurologic...).

The interior stars — shared the mechanisms (genes, pathways, metabolites) where diseases converge.

The — **hypotheses**, each *disease* → **The** — **void the**; a thread through
braids the running *shared* **empty** the **engine** the middle would
centre **never** be a generic
→ *partner* **crosses** shortcut.
disease

Position = quality. Specific, expert-ready discoveries sit near the bright **rim**; generic scale-free hubs collapse toward the **centre**. The geometry *is* the ranking.

02 Hover a disease on the ring

Move the cursor over a disease name on the outer ring



Hovering a disease lights every hypothesis it takes part in.

What it shows: every hypothesis that disease participates in — the interior mechanism-nodes it routes through, and the partner diseases it links to. The tooltip reads its **clinical family** and “**N links**” — how many hypotheses involve it.

Why it matters: a disease with many links is a well-connected node in the convergence network; follow its brightest threads to the mechanisms it shares most strongly.

03 Hover a star (interior node)

Hover one of the glowing interior points (a GO / Reactome / CHEBI node)



Every star is a shared biological mechanism; the tooltip carries its physics.

What it shows: all hypotheses that converge on that mechanism light up, and the tooltip gives you the node's real numbers — **best CQS** (quality of the best hypothesis through it), **r** (its Poincaré radius: closer to 1 = specific, closer to 0 = generic), and **how many hypotheses converge here**.

Why it matters: this is where the data lives. A node with a high count *and* a high **r** is a **specific shared bottleneck** — a genuine discovery. A high count with low **r** is a generic hub to treat with caution.

04 Hover & click a candidate

Hover a card in “Priority Candidates” to highlight — click it to open the full dossier



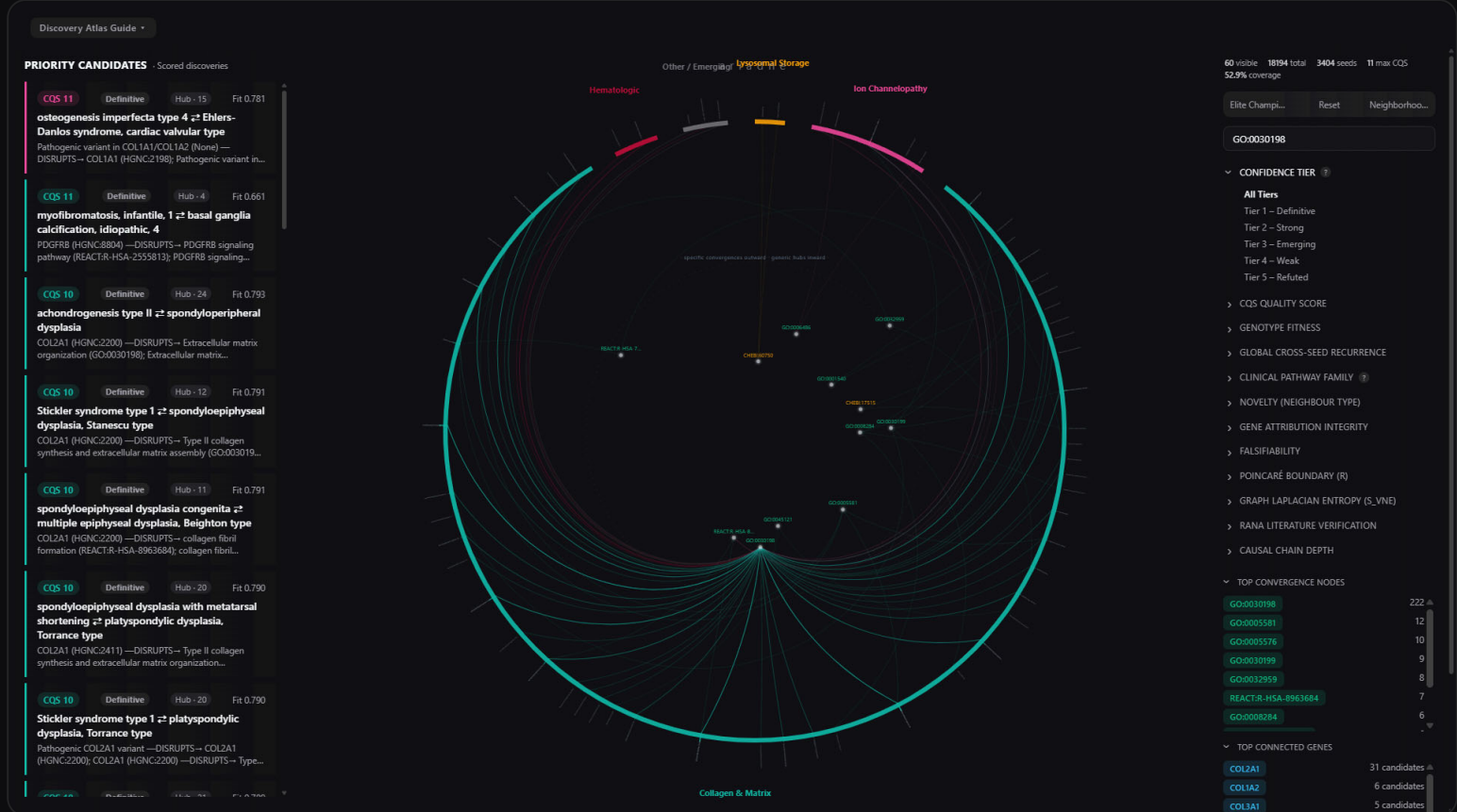
One pair, plus its family of related hypotheses.

Hover a card and its braid lights **brightest**, while every **sibling hypothesis that shares the same convergence node** lights more softly — so you see the pair *and* the mechanistic neighbourhood it belongs to.

Click the card to open the candidate's **full discovery dossier** — the complete, expert-ready report (see §11).

05 Click a Top Convergence Node

Open “Top Convergence Nodes” and click a **GO:...** / **REACT:...** / **CHEBI:...** chip



Isolating a single biological failure point across the whole corpus.

What the number means: the value beside each node (e.g., **GO:0030198 – 193**) is **how many candidate hypotheses route their mechanism through that node**. Clicking it filters the Atlas to **every disease pair that converges on that exact biological point**.

Why it matters: these are the **shared bottlenecks of rare disease** — the specific molecular steps where many separate conditions fail. A tight, specific node (small pathway, high r) that many diseases share is precisely what an expert wants to see.

06 Click a Top Connected Gene

Open “Top Connected Genes” and click a symbol (e.g., **COL2A1** , **PTPN11** , **KRAS**)



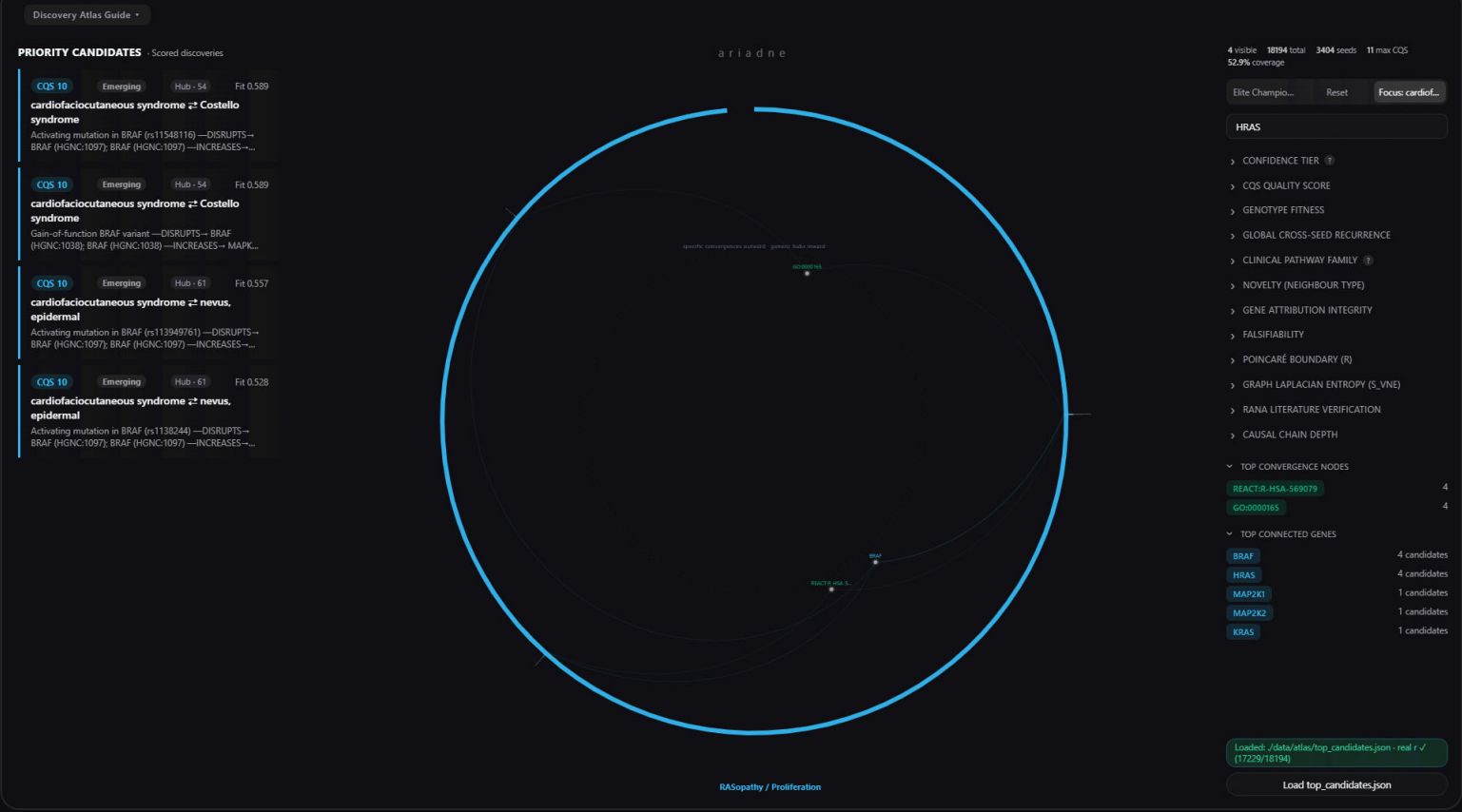
A single gene bridging several clinical families.

What the number means: “ COL2A1 – 109 candidates ” = the gene appears in the mechanism of **109 candidate hypotheses**. Clicking it lights every disease pair that converges on that gene.

Why it matters — the key discovery signal: if a gene lights pairs across **many different families** (its threads reach several coloured arcs), it is a **pleiotropic, cross-domain bridge** — one molecular root behind diseases that look clinically unrelated. That is one of the most valuable things ARIADNE can surface. *Caveat:* check where its stars sit — a bridge on the specific **rim** is a discovery; a promiscuous hub collapsed at the **centre** (a TP53 / KRAS -type connector) is expected background.

07 Neighbourhood Focus Mode

Click “Neighborhood Focus Mode”

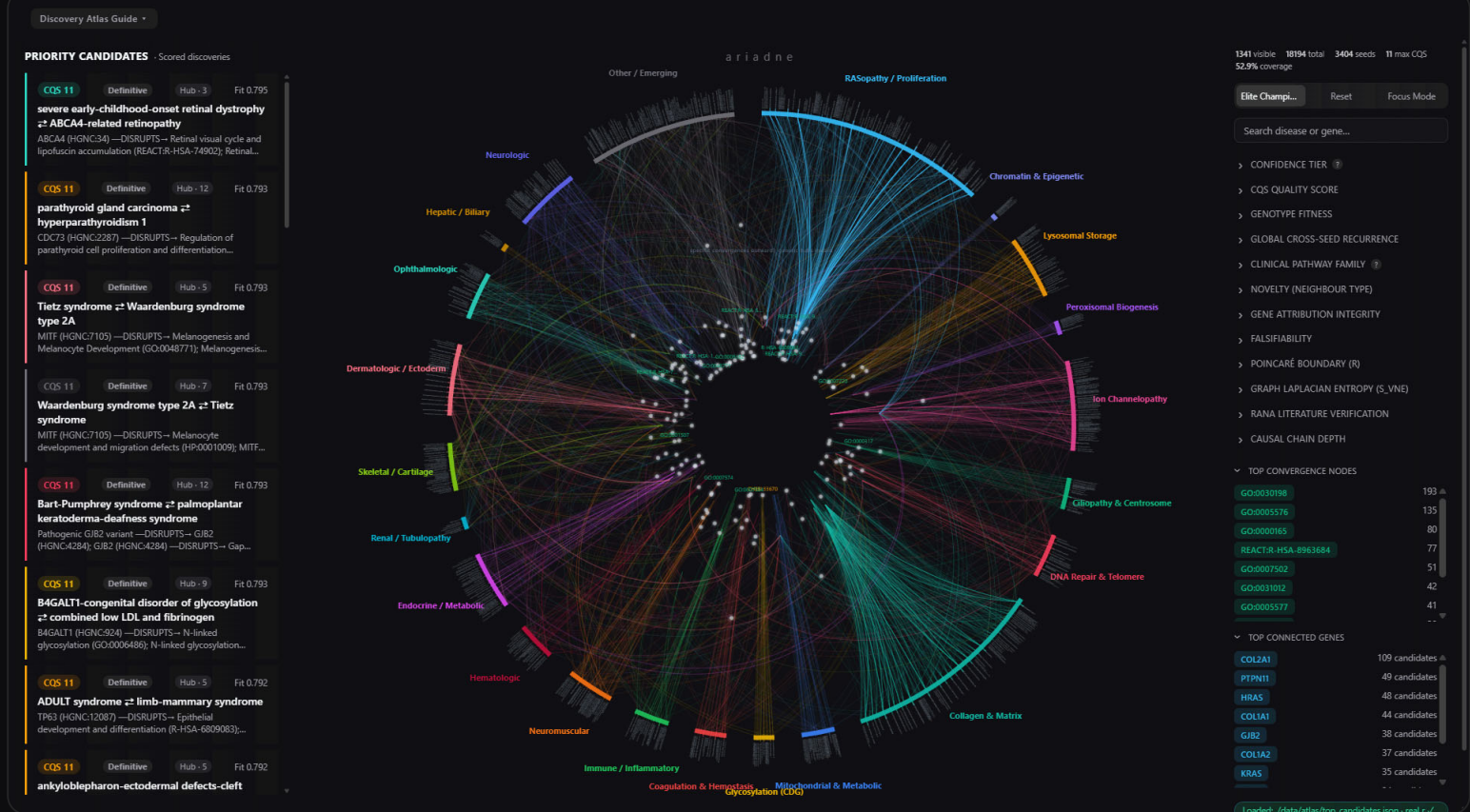


Everything except one disease's neighbourhood falls away.

- What it does:** isolates the top candidate's **seed disease** and shows only its neighbourhood — every mechanism and partner it connects to. The button becomes `Focus: <disease> (Exit)` ; click again to leave.
- Why it matters:** use it to study one disease in depth — a clean, uncluttered view of everything ARIADNE thinks it shares mechanism with.

08 Elite Champions

Click “Elite Champions”



The strongest, most-corroborated candidates only.

What it does: keeps only the highest-quality hypotheses — **fitness ≥ 0.65 , CQS ≥ 7 , and independent recurrence across ≥ 3 seed runs** (Hub ≥ 3), on a reasonably specific pathway. Everything weaker is hidden.

Why it matters: this is your “show me only what's worth an expert's time” switch — the shortlist you'd bring to a domain scientist.

09 The physics filters — r and S_VNE

ARIADNE places every candidate in **hyperbolic space** and measures its **graph thermodynamics**. Two filters expose that directly.



Poincaré filter set to the specific rim — the discoveries.



Graph Laplacian entropy (S_{VNE}) — a mechanism's structural complexity.

POINCARÉ BOUNDARY (R)

r is a candidate's distance from the centre in hyperbolic space, driven by how *specific* its shared

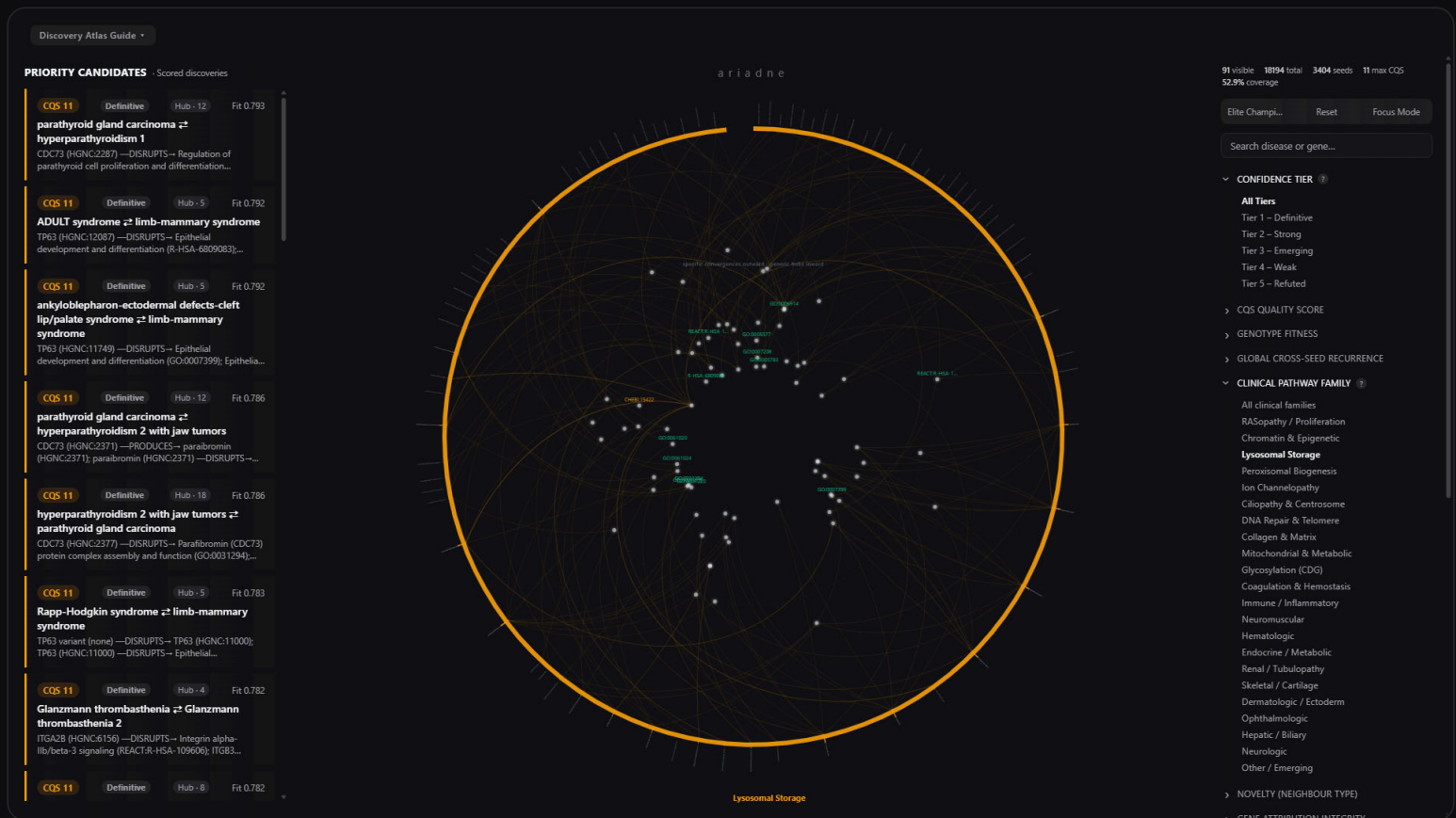
pathway is. $r \geq 0.50$ = the **specific rim** (rare, expert-ready discoveries, ~4% of the corpus). $r < 0.20$ = **generic hubs** at the core. Use it to strip the diagram down to only the specific convergences.

GRAPH LAPLACIAN ENTROPY (S_{VNE})

S_{VNE} is the spectral entropy of a hypothesis's causal graph — a measure of its structural complexity. Tightly organised, focused mechanisms have low entropy; sprawling, diffuse ones have high entropy. It's the thermodynamic fingerprint behind the $\kappa_t = a \cdot e^{-b \cdot S_{VNE}}$ law (fits at $R^2 = 0.94$ across the corpus).

10 The 21 clinical families

Watch the coloured rim arcs — or filter to one family



A single family and the mechanisms it converges on.

What it shows: every disease is coloured by its **clinical family**, inferred from the *mechanism* it converges on (not just its name). Threads that jump between differently-coloured arcs are **cross-domain bridges** — the non-obvious links ARIADNE exists to find.

Why it matters: the more a hypothesis bridges *different* families, the more surprising — and the higher ARIADNE scores it (the cross-domain CQS bonus). Cross-family threads are where discovery lives; same-family threads are usually calibration.

11 The discovery dossier

CQS Score: 11 | Tier: 3: Emerging | Global Hub (6 seeds) | Fitness: 0.564

hypomyelinating leukodystrophy 3 ± Charcot-Marie-Tooth disease recessive intermediate B

Also prepared by seeds: MONDO:0014506, MONDO:0014116, MONDO:0013489, MONDO:0014335, Charcot-Marie-Tooth disease type 2D

Key genes appear in 35 other candidates across the Atlas

Shared Gene-Sets Pathways (3):

- Reactome:R-HSA-1403021
- Reactome:R-HSA-372736
- Reactome:R-HSA-3866426

PROPOSED MECHANISM FLOW

```

KARS1 (HGNC:3545) --disrupts--> aminocyl-tRNA synthetase complex assembly and function (GO:0005054)
aminocyl-tRNA synthetase complex assembly and function (GO:0005054) --leads to--> Leukodystrophy (H0002214)
aminocyl-tRNA synthetase complex assembly and function (GO:0005054) --leads to--> Axonal neuropathy (H0004676)
  
```

Loss-of-function mutations in KARS1 disrupt the function and assembly of the aminocyl-tRNA synthetase complex, leading to distinct but related neurological phenotypes: leukodystrophy in hypomyelinating leukodystrophy 3 and axonal neuropathy in Charcot-Marie-Tooth disease recessive intermediate B.

[Show raw mechanism map](#)

Falsification Test: The discovery of distinct, non-overlapping molecular pathways downstream of KARS1 mutations in each disease, or identification of a different causal gene for one of the diseases, would falsify this hypothesis. Additionally, finding pathogenic variants in KARS1 with a high allele frequency (>0.01) in gnomAD would suggest it is not a rare disease gene.

[View Raw Discovery Log JSON](#) | [Copy Dossier](#) | [Download Markdown](#)

discovery_MONDO_0008441_2026-07-20T16:12:27-0542.json

FULL DISCOVERY RUN FOR HYPOMYELINATING LEUKODYSTROPHY 3

#	Partner Disease	Fitness	Shared Pathways	Key Gene(s)
1	Charcot-Marie-Tooth disease recessive intermediate B	0.564	3	KARS1
2	diffuse cerebral and cerebellar atrophy - intractable seizures - progressive microcephaly syndrome	0.549	3	QARS1
3	hypomyelination with brain stem and spinal cord involvement and leg spasticity	0.546	3	AIMP1, DARS1
4	hypomyelinating leukodystrophy 9	0.539	3	AIMP1, KARS1
5	leukodystrophy, hypomyelinating, 15	0.539	3	AIMP1, AIMP1, EPRS1
6	growth retardation, intellectual developmental disorder, hypotonia, and hepatopathy	0.537	3	IARS1
7	infertile liver failure syndrome 1	0.537	3	IARS1
8	deafness, congenital, and adult-onset progressive leukoencephalopathy	0.537	3	KARS1
9	autosomal recessive nonsyndromic hearing loss 89	0.486	3	KARS1, KARS1
10	leukodystrophy, hypomyelinating, 17	0.476	3	AIMP1, AIMP2

The complete, expert-ready report behind a single hypothesis.

What it shows: the full case for the pair — the typed causal chain (*Variant* → *Gene* → *Pathway* → *Metabolite* → *Phenotype*), the shared genes and pathways, the metrics (CQS, fitness, tier, cross-seed recurrence), and the **falsification test:** the exact experiment that would disprove it.

Why it matters: this is what you hand an expert. **Copy Dossier** or **Download Markdown** to share it. Every claim traces back to an authoritative database, and every hypothesis states how it could be proven wrong — science, not storytelling.

How to read a discovery in one glance

- **Bright thread reaching the rim** → a specific, high-quality convergence. Worth reading.
- **A star with high count + high r** → a specific mechanism many diseases genuinely share. A bottleneck of rare disease.
- **A gene lighting several families** → a pleiotropic cross-domain bridge — one root, many clinical faces. The strongest discovery signal (if it sits toward the rim).
- **Everything collapsed at the centre** → generic scale-free hubs. Expected background, demoted on purpose.

- **A cross-family braid** → a non-obvious bridge; exactly what the engine is built to surface.

The one sentence: you're not looking at a diagram of data — you're looking at a map of **ranked, testable predictions**. Click any thread to read its step-by-step mechanism and the experiment that would prove it wrong.

ARIADNE — a genetic algorithm whose organisms are scientific hypotheses. · Discovery Atlas Field Guide